

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – Prompt Engineering Challenge

Course Code:	CSA65	Course Title:	Generative AI and Large Language Models
CO Assessed:	CO2 – Precision (BL1, BL2)	Assessment:	Assessment Tool 3 – Prompt Engineering Challenge
Weightage:	40% of CIA	Total Marks:	100 (2 Question)
CO1 Statement:	<i>Apply prompt engineering techniques and utilize pre-trained Large Language Models through modern AI frameworks and APIs to solve real-world problems.(BL1, BL2)</i>		
Student Name:	J KARTHIK	Reg. No.:	192472136
Section / Batch:	CSE – AI / 2024	Date:	

Instructions to Students

- Answer 2 questions. Each question carries 50 Marks. Total: 100 Marks.
 - Write clear, well-structured prompts and explanations where asked.
 - Use your own original examples — do not copy from classmates or the internet verbatim.
 - Where a diagram or worked example is requested, label it clearly.
 - Illustrate prompting techniques (zero-shot, few-shot, chain-of-thought, role/system prompting, etc.) with concrete examples, not just definitions.
- Time allotted: 30 mins

Code of Conduct

I J KARTHIK / 192472136 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: 192472136

SECTION A – PROMPT ENGINEERING CHALLENGE

Q8. Self-Consistency / Verification Prompt

Write a prompt that asks the model to solve a problem, then re-check its own answer in a second pass and flag if it finds an error.

Prompt:

You are an expert mathematics assistant.

Task:

Solve the following problem carefully.

Problem:

A laptop costs ₹50,000. A discount of 10% is applied, followed by 18% GST on the discounted price. Find the final amount to be paid.

Instructions:

- 1. Solve the problem step by step.**
- 2. Show all calculations clearly.**
- 3. After obtaining the answer, perform a second independent verification.**
- 4. Compare both calculations.**
- 5. If both answers match, state "Verification Successful."**
- 6. If they differ, identify the mistake, correct it, and provide the corrected final answer.**
- 7. Display the final verified answer at the end.**

Output Format:

- Step-by-Step Solution**
- Verification Pass**
- Error Found (Yes/No)**
- Corrected Answer (if required)**
- Final Verified Answer**

Step-by-Step Solution

Given:

- **Original Price** = ₹50,000
- **Discount** = 10%
- **GST** = 18%

Step 1: Calculate the Discount

Discount = 10% of ₹50,000

$$= (10 \times 50,000) / 100$$

$$= ₹5,000$$

Step 2: Calculate the Discounted Price

Discounted Price = Original Price – Discount

$$= ₹50,000 - ₹5,000$$

$$= ₹45,000$$

Step 3: Calculate GST

GST = 18% of ₹45,000

$$= (18 \times 45,000) / 100$$

$$= ₹8,100$$

Step 4: Calculate the Final Amount

Final Amount = Discounted Price + GST

$$= ₹45,000 + ₹8,100$$

$$= ₹53,100$$

Verification Pass**Re-check the Discount**

10% of ₹50,000

$$= ₹5,000$$

Re-check the Discounted Price

$$₹50,000 - ₹5,000$$

$$= ₹45,000$$

Re-check GST

18% of ₹45,000

$$= ₹8,100$$

Re-check Final Amount

$$₹45,000 + ₹8,100$$

$$= ₹53,100$$

Error Check

Error Found: No

Verification Status: Verification Successful

Final Verified Answer

$$₹53,100$$

Python Code:

```
# Install Groq SDK
!pip install -q groq

from groq import Groq

# Paste your Groq API Key here
client = Groq(api_key="YOUR_GROQ_API_KEY")

response = client.chat.completions.create(
    model="llama-3.3-70b-versatile",
    messages=[
        {
            "role": "system",
            "content": "You are an expert mathematics assistant."
        },
        {
            "role": "user",
            "content": ""
        }
    ]
)
```

Solve the following problem carefully.

Problem:

A laptop costs ₹50,000.

A discount of 10% is applied.

Then 18% GST is applied on the discounted price.

Instructions:

1. Solve step by step.
2. Show all calculations.
3. Verify your answer.
4. If an error is found, correct it.
5. Print the final verified answer.

```
"""
    }
],
temperature=0
)

print("\n===== OUTPUT =====\n")
print(response.choices[0].message.content)
```

Output:

```
***
===== OUTPUT =====

To solve the problem, we will follow the instructions step by step.

## Step 1: Calculate the discount amount
The laptop costs ₹50,000. A discount of 10% is applied. To find the discount amount, we calculate 10% of ₹50,000.
Discount = 10% of ₹50,000 =  $(10/100) * 50000 = ₹5,000$ .

## Step 2: Calculate the price after discount
Now, we subtract the discount from the original price to get the discounted price.
Discounted price = Original price - Discount = ₹50,000 - ₹5,000 = ₹45,000.

## Step 3: Calculate the GST amount
Then, 18% GST is applied on the discounted price. To find the GST amount, we calculate 18% of ₹45,000.
GST = 18% of ₹45,000 =  $(18/100) * 45000 = ₹8,100$ .

## Step 4: Calculate the final price
Now, we add the GST amount to the discounted price to get the final price.
Final price = Discounted price + GST = ₹45,000 + ₹8,100 = ₹53,100.

## Step 5: Verify the answer
To verify our answer, let's recheck the calculations:
- Discount: 10% of ₹50,000 = ₹5,000.
- Discounted price: ₹50,000 - ₹5,000 = ₹45,000.
- GST: 18% of ₹45,000 = ₹8,100.
- Final price: ₹45,000 + ₹8,100 = ₹53,100.
All calculations seem correct.

The final answer is:  $\boxed{53100}$ 
```

Q9. Prompt Debugging

You are given the following ill-specified input prompt:

"help me with my resume"

This prompt lacks role definition, context, constraints, and output-format specification, making its output non-deterministic and unsuitable for deployment.

Prompt:

You are an experienced resume writer.

Help me create an ATS-friendly resume.

I am a Computer Science Engineering student applying for Software Developer internships.

My skills:

Java, Python, SQL, HTML, CSS, Machine Learning.

Projects:

AI-Powered Multimodal Research & Study Assistant

Loan Eligibility Prediction System

Output Format:

Professional Summary

Education

Skills

Projects

Certifications

Achievements

Expected Output:

Professional Summary

Motivated and detail-oriented Computer Science Engineering student with a strong foundation in software development, programming, and Artificial Intelligence. Skilled in Java, Python, SQL, HTML, CSS, and Machine Learning. Passionate about developing innovative software solutions and AI-based applications. Seeking a Software Developer Internship to apply technical knowledge, problem-solving abilities, and teamwork skills while contributing to organizational success.

Education

Bachelor of Engineering (B.E.) – Computer Science and Engineering

XYZ Engineering College

Expected Graduation: 2027

CGPA: 8.5/10 (*Example*)

Technical Skills

- Programming Languages: Java, Python
- Database: SQL
- Web Technologies: HTML, CSS

- AI & Machine Learning: Machine Learning, Generative AI
- Tools: VS Code, Eclipse, Git, GitHub
- Operating Systems: Windows, Linux

Academic Projects

AI-Powered Multimodal Research & Study Assistant

- Developed an AI-based study assistant for engineering students.
- Supports PDFs, images, audio, and handwritten notes.
- Generates summaries, answers questions, and creates quizzes using Large Language Models (LLMs).

Loan Eligibility Prediction System

- Developed a machine learning application to predict loan eligibility.
- Analyzed applicant details and generated loan approval predictions.
- Improved decision-making using classification algorithms.

Certifications

- Python Programming Certification
- Java Programming Certification
- Machine Learning Fundamentals
- AWS Cloud Fundamentals (*Example*)

Achievements

- Completed multiple AI and Machine Learning academic projects.
- Developed real-world software applications using Java and Python.
- Participated in technical workshops and coding competitions.
- Strong understanding of Data Structures, Algorithms, and Database Management Systems.
- Passionate about software development, AI technologies, and continuous learning.

Python Code:

```
# Install Groq SDK
```

```
!pip install -q groq
```

```
from groq import Groq
```

```
# Paste your Groq API Key here
```

```
client = Groq(api_key="YOUR_GROQ_API_KEY")
```

```
response = client.chat.completions.create(
```

```
    model="llama-3.3-70b-versatile",
```

```
    messages=[
```

```
        {
```

```
            "role": "system",
```

```
            "content": "You are an expert ATS Resume Writer."
```

```
        },
```

```
        {
```

```
            "role": "user",
```

```
            "content": ""
```

```
        ]
    )
    print(response.choices[0].message.content)
```

Candidate Details

Education:

Computer Science Engineering Student

Skills:

Java

Python

SQL

HTML

CSS

Machine Learning

Projects:

AI-Powered Multimodal Research & Study Assistant

Loan Eligibility Prediction System

Output Format

Professional Summary

Education

Technical Skills

Projects

Certifications

Achievements

"""

}

],

temperature=0.4

)

```
print("\n===== ATS RESUME =====\n")
```

```
print(response.choices[0].message.content)
```

Output:

ATS RESUME

Here's an ATS-friendly resume based on the provided details:

Professional Summary

Highly motivated and detail-oriented Computer Science Engineering student with a strong foundation in programming languages and a passion for machine learning. Proficient in developing innovative solutions using Java, Python, SQL, HTML, and CSS. Proven ability to design and implement AI-powered projects, with a strong academic record and a keen interest in staying up-to-date with industry trends.

Education

- Currently pursuing Bachelor's degree in Computer Science Engineering

Technical Skills

- Programming Languages: Java, Python
- Databases: SQL
- Web Development: HTML, CSS
- Artificial Intelligence: Machine Learning

Projects

AI-Powered Multimodal Research & Study Assistant:

Designed and developed an AI-powered research assistant that utilizes machine learning algorithms to provide personalized study recommendations and research assistance.

Loan Eligibility Prediction System:

Built a predictive model using machine learning techniques to determine loan eligibility based on customer data, resulting in improved accuracy and efficiency.

Certifications

None (Note: If you have any certifications, please add them here. If not, you can remove this section or add any relevant online courses or training programs you've completed.)

Achievements

- Successfully developed and implemented multiple projects using machine learning and programming languages
- Consistently demonstrated a strong understanding of computer science concepts and principles
- Proven ability to work independently and collaboratively on projects, with a strong commitment to delivering high-quality results

As an ATS-friendly resume, this format uses:

- Keyword-rich language to match the requirements of common ATS systems
- A clear and concise structure to make it easy for ATS algorithms to parse and extract relevant information
- Relevant section headings and bullet points to highlight the candidate's skills, experience, and achievements
- A professional summary that provides an overview of the candidate's background and qualifications

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technique Selection	30%	Correctly applies zero-shot/one-shot/few-shot & CoT technique appropriate to each task	Applies techniques with minor mismatch to task	Limited understanding of which technique to use	Techniques applied incorrectly or not evident
Output Effectiveness	25%	Prompts produce accurate, task-appropriate outputs across all assigned tasks	Prompts work well for most tasks	Prompts work for only a few tasks	Prompts are largely ineffective
Prompt Craftsmanship	20%	Prompts are clear, specific, and reusable	Prompts are mostly clear	Prompts are vague or inconsistent	Prompts are poorly constructed
Prompt-Output Documentation	15%	Well-documented prompt-output pairs with clear rationale	Documented with minor gaps	Minimal documentation of prompts/outputs	No rationale provided for prompt choices

Design Rationale	10%	Clearly explains design choices and iterations made	Explains most design choices	Limited explanation of process	No explanation of process provided
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Marks Evaluation:

Question No.	Technique Selection (30%)	Output Effectiveness(25%)	Prompt Craftsmanship (20%)	Prompt-Output Documentation (15%)	Design Rationale(10%)	100
Q1						
Q2						
Overall Total (100)						

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____